

IDEATION PHASE

Metro Ticket Generating System in ServiceNow

Empathize & Discover • Brainstorm & Prioritize • Define Problem Statements

Date	15 September 2026
Name	Murala Ayyappa
Roll Number	23481A4268
Project Name	Metro Ticket Generating System in ServiceNow

1. Empathy Map Canvas

Empathy Map Canvas — Passenger / Metro Commuter Perspective

An empathy map captures the user's behaviours, attitudes, goals, concerns, and challenges. For this project, the primary persona considered during ideation is a metro passenger who wants to book a ticket quickly and conveniently.

THINK & FEEL • Wants a quick and simple ticket-booking process. • Wants confidence that the fare is calculated correctly. • Wants a convenient digital payment option. • Wants a ticket that can be easily presented at the metro gate. • Feels frustrated when booking takes too long or requires repeated manual steps.	HEAR • Friends and fellow passengers expect faster digital ticketing. • Passengers discuss the need for convenient cashless payment. • Users expect modern metro services to provide digital tickets and QR codes. • People value clear fare information before completing a booking.
SEE • Queues and crowded ticket counters during busy periods. • Different source and destination stations. • Existing digital payment methods such as UPI and cards. • A need for a clear confirmation/ticket after booking. • Digital QR codes can be used as a convenient ticket representation.	SAY & DO • Selects starting and destination stations. • Chooses single or return journey. • Selects number of passengers. • Checks the calculated amount. • Selects a payment mode and provides additional payment information when required. • Uses the generated ticket/QR code for the journey.
PAIN • Long or slow manual ticket-booking process. • Difficulty calculating the correct fare for different routes. • Need to carry or handle cash when digital payment is unavailable. • Risk of confusion when booking passenger counts or journey type. • Lack of an immediately accessible digital ticket.	GAIN • Fast and convenient digital booking. • Automatic fare calculation. • Multiple payment options. • QR-based digital ticket. • Consistent ticket/request records in ServiceNow. • Reduced manual effort and better booking experience.

Empathy Map Summary

Primary User	Metro passenger / commuter
Main Goal	Book a metro ticket quickly, accurately, and conveniently.
Main Frustrations	Slow/manual booking, fare uncertainty, payment inconvenience, and lack of an accessible digital ticket.
Expected Benefits	Automated fare calculation, digital payment choices, QR ticket generation, and reliable booking records.

2. Brainstorm & Idea Prioritization

Brainstorming provides a free and open environment for generating creative solutions. The supplied template organizes this activity into three steps: team gathering and problem selection, idea listing/grouping, and idea prioritization.

Step 1: Team Gathering, Collaboration and Select the Problem Statement

Selected design challenge:

How might we make metro ticket booking faster, simpler, and more reliable by automating station selection, fare calculation, payment information, QR ticket generation, and ticket record management in ServiceNow?

Step 2: Brainstorm, Idea Listing and Grouping

Idea / Concept	Grouping	Description
Digital Ticket Booking	Booking Experience	Service Catalog item for selecting source, destination, journey type, and passenger count.
Automatic Fare Calculation	Fare & Validation	Calculate the fare from the selected source/destination and journey type.
Return Journey Support	Fare & Validation	Calculate return fare automatically for return journeys.
Multiple Passenger Support	Booking Experience	Allow passengers to select the required number of passengers.
Multiple Payment Modes	Payment	Provide UPI, Card, and Others as payment choices.
Conditional Payment Details	Payment	Show an additional payment-mode field only when Others is selected.
QR Ticket Generation	Digital Ticket	Generate a QR-based digital ticket after booking.
Metro Station Master Data	Data Management	Maintain metro station details in a dedicated ServiceNow table.
Central Ticket Database	Data Management	Store booking information, fare, payment mode, journey details, and QR status.
RITM Tracking	ServiceNow Integration	Maintain a readable Requested Item number with the metro ticket record.
Validation	Fare & Validation	Prevent selecting the same starting and destination station.
Automated Record Creation	ServiceNow Automation	Use Flow Designer to create a Metro Database record from the catalog request.

Step 3: Idea Prioritization

Ideas are prioritized using the supplied template's two dimensions: Importance and Feasibility. The following prioritization reflects the project needs and the functionality implemented in the ServiceNow solution.

Idea	Importance	Feasibility	Priority
Automatic Fare Calculation	High	High	Critical
Digital Ticket Booking	High	High	Critical
QR Ticket Generation	High	High	Critical
Central Ticket Database	High	High	Critical
Automated Record Creation	High	High	Critical
Metro Station Master Data	High	High	Critical
Multiple Payment Modes	High	High	High
Return Journey Support	High	High	High
Multiple Passenger Support	High	High	High
Validation	High	High	High
Conditional Payment Details	Medium	High	Medium
RITM Tracking	Medium	High	Medium

Prioritized Outcome

- First priority: provide a reliable digital booking flow with automatic fare calculation and automated record creation.
- Second priority: provide QR-based digital ticketing and complete ticket information in the Metro Database.
- Third priority: support payment choices, return journeys, multiple passengers, and validation.
- Supporting priority: maintain readable RITM tracking and conditional payment information.

3. Define the Problem Statements

The supplied Customer Problem Statement template uses the structure: I am (Customer) → I'm trying to → But → Because → Which makes me feel. The statements below translate the passenger's needs and pain points into project-focused problem statements.

PS-1

I am (Customer)	Customer and attributes	a metro passenger who needs to travel between stations
I'm trying to	Outcome / job to be done	book a metro ticket quickly with the correct fare
But	Problems / barriers	manual or slow ticket booking can require extra time and effort
Because	Root cause / reason	the fare, passenger count, journey type, and booking details may need to be handled accurately
Which makes me feel	Emotional impact	frustrated and uncertain about whether the booking and fare are correct.

PS-2

I am (Customer)	Customer and attributes	a metro passenger who prefers convenient digital services
I'm trying to	Outcome / job to be done	complete a ticket booking using a suitable payment mode
But	Problems / barriers	payment options and additional payment information may not be presented in a simple way
Because	Root cause / reason	different passengers may prefer UPI, card, or another payment method
Which makes me feel	Emotional impact	inconvenienced when the payment process is unclear or restrictive.

PS-3

I am (Customer)	Customer and attributes	a metro passenger who needs a usable ticket after booking
I'm trying to	Outcome / job to be done	receive and use a digital ticket easily
But	Problems / barriers	a physical/manual ticket process can make ticket access and verification less convenient
Because	Root cause / reason	a digital QR representation can provide a simple way to present ticket information
Which makes me feel	Emotional impact	concerned about carrying, accessing, or presenting the ticket when needed.

Final Ideation Problem Statement

A metro passenger needs a simple, fast, and reliable way to book a metro ticket with accurate fare calculation, convenient payment choices, and an accessible digital QR ticket. The Metro Ticket Generating System in ServiceNow addresses this need by combining Service Catalog booking, automated fare logic, conditional payment information, QR generation, and automated ticket record creation in a centralized ServiceNow database.

Ideation Phase Completion Checklist

- ☐ Empathy Map Canvas completed for the metro passenger persona.
- ☐ Passenger Think & Feel, Hear, See, Say & Do, Pain, and Gain identified.
- ☐ Problem/design challenge selected for brainstorming.
- ☐ Ideas generated and grouped by booking, fare, payment, digital ticket, data, and ServiceNow automation themes.
- ☐ Ideas prioritized using Importance and Feasibility.
- ☐ Customer problem statements documented using the supplied I-am / trying-to / but / because / feel structure.
- ☐ Final ideation problem statement aligned with the Metro Ticket Generating System project.